

TRAIL VALLEY, NT, Canada

WMO: 716830

Lat: 68.742N

Lon: 133.499W

Elev: 279

StdP: 14.55

Time Zone: -7.00 (NAM)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.8	-32.6	-42.3	0.5	-35.4	-38.9	0.6	-32.3	40.9	-0.9	35.8	-0.9	4.3	0	0.910

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	76.2	60.1	72.1	58.8	68.0	57.4	62.1	72.6	60.1	69.3	58.0	66.3	10.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.0	72.5	65.4	55.9	67.1	63.4	53.9	62.4	61.6	28.1	72.8	26.6	69.7	25.3	66.2	70.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.5	23.0	19.7	-40.5	82.9	3.5	3.6	-43.0	85.4	-45.1	87.5	-47.0	89.5	-49.6	92.1		
			-40.1	65.9	3.9	2.0	-43.0	67.4	-45.3	68.5	-47.5	69.7	-50.4	71.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.2	-11.3	-10.0	-8.3	8.9	30.5	48.4	54.5	49.4	38.9	21.7	1.4	-8.3	
	DBStd	26.82	12.99	13.23	12.39	12.78	11.96	9.22	7.55	7.84	7.55	8.74	10.28	12.25	
	HDD50	12002	1899	1679	1807	1232	614	139	40	107	342	877	1458	1808	
	HDD65	17112	2364	2099	2272	1682	1071	501	333	486	782	1341	1908	2273	
	CDD50	381	0	0	0	0	8	92	180	90	10	1	0	0	
	CDD65	17	0	0	0	0	0	4	9	4	0	0	0	0	
	CDH74	218	0	0	0	0	7	77	99	35	0	0	0	0	
	CDH80	32	0	0	0	0	1	15	12	4	0	0	0	0	
Wind	WSAvg	8.4	8.8	9.1	8.4	8.0	8.8	9.4	9.2	8.7	8.6	7.7	7.3	7.2	
Precipitation	PrecAvg	9.4	0.4	0.4	0.5	0.4	0.5	1.0	1.6	1.6	1.2	0.9	0.6	0.5	
	PrecMax	12.4	1.2	0.8	1.3	1.0	1.4	2.7	2.9	3.2	2.2	1.7	1.3	1.4	
	PrecMin	7.1	0.0	0.0	0.0	0.0	0.0	0.1	0.4	0.5	0.3	0.2	0.0	0.1	
	PrecStd	1.4	0.3	0.2	0.3	0.3	0.3	0.7	0.6	0.6	0.5	0.4	0.3	0.3	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.4	26.9	31.7	43.5	69.2	82.3	81.6	78.8	65.7	47.6	31.4	25.8	
		MCWB	30.0	25.5	29.8	38.5	52.6	62.3	62.5	63.8	53.6	42.8	30.9	25.0	
	2%	DB	22.9	21.3	22.0	38.0	61.3	75.2	76.6	71.9	59.4	39.0	24.7	20.1	
		MCWB	21.9	20.1	20.7	34.4	48.4	58.9	60.2	60.4	50.3	36.9	23.9	19.5	
	5%	DB	14.7	16.3	15.0	33.3	54.0	70.7	72.9	66.9	54.8	35.6	20.5	13.9	
		MCWB	14.0	15.4	13.7	30.7	45.1	56.3	59.8	58.3	48.6	34.0	19.8	13.4	
	10%	DB	6.7	10.3	9.6	28.6	47.5	65.6	69.1	62.7	50.3	32.8	16.4	8.9	
		MCWB	6.1	9.6	8.4	26.7	41.5	54.1	58.6	55.9	46.3	31.8	15.7	8.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.2	25.3	30.0	36.2	52.0	63.4	64.6	64.4	54.8	45.1	31.1	25.4
			MCDB	29.2	26.3	32.3	39.4	65.1	78.0	74.8	76.1	61.7	49.2	31.7	26.3
2%		WB	21.9	20.3	21.6	32.5	47.7	59.9	62.6	61.5	51.8	37.6	24.8	19.7	
		MCDB	22.9	21.3	22.8	35.2	58.8	72.8	73.0	70.2	57.7	39.2	25.3	20.3	
5%		WB	14.9	15.9	14.8	29.4	44.2	57.4	60.7	59.0	49.0	34.6	20.6	13.4	
		MCDB	15.5	16.8	16.1	31.6	52.2	69.3	70.0	64.9	54.1	35.7	21.3	14.0	
10%		WB	7.3	10.0	8.9	25.4	40.8	54.7	58.9	56.3	46.5	32.1	16.6	8.8	
		MCDB	7.8	10.7	10.1	27.1	46.0	64.5	67.4	61.7	50.6	33.1	17.2	9.2	
Mean Daily Temperature Range	5% DB	MDBR	11.3	12.6	13.3	15.4	15.3	19.5	17.5	14.9	11.7	9.2	10.6	11.0	
		MCDBR	18.4	15.9	16.0	16.6	25.9	27.0	24.4	21.8	19.3	11.1	11.9	16.6	
		MCWBR	17.1	15.0	14.5	14.3	15.4	13.9	12.1	12.0	11.1	9.1	11.6	16.3	
	5% WB	MCDBR	17.8	15.5	15.6	16.2	24.0	25.4	21.7	19.9	18.4	10.9	11.8	16.9	
		MCWBR	17.1	14.9	14.4	13.7	14.7	13.9	12.0	11.7	11.3	8.9	11.7	16.7	
Clear-Sky Solar Irradiance	taub	0.157	0.212	0.250	0.302	0.336	0.338	0.360	0.334	0.298	0.233	0.157	N/A		
	taud	1.908	2.111	2.221	2.173	2.315	2.457	2.439	2.520	2.545	2.217	1.917	N/A		
	Ebn at Noon	29	189	252	266	270	273	262	258	239	175	23	0		
	Edh at Noon	2	13	23	33	33	30	29	24	18	12	2	0		
All-Sky Solar Radiation	RadAvg	9	155	673	1402	1854	1970	1590	1063	569	183	24	0		
	RadStd	1	11	32	57	129	151	148	109	52	21	2	0		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (7 neighbors)		+1.87	+3.26	+3.69	N/A	N/A	N/A	-769	-760	N/A	N/A	

Nomenclature: See separate page